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**NexChain**

# Supply Chain Intelligence Co- Pilot

# The Problem

## **Scattered Supply Chain Data Challenges**

Supply chain data is fragmented across various systems, causing teams to waste time searching for answers, leading to inefficiencies and potential delays in operations.



# Key Stakeholders

Identifying those affected by inefficiencies

## Supply Chain Manager

The Supply Chain Manager requires **instant answers** on order delays and next steps, ensuring timely decision-making and minimizing disruptions in the supply chain.

## Customer Service Rep

Customer Service Representatives need **clear, plain-language** answers for customers, facilitating quick responses to inquiries about order statuses without dealing with complex data.

## Operations Analyst

Operations Analysts seek **quick access** to inventory and warehouse reports, eliminating the need for complex SQL queries, thus enhancing productivity and operational efficiency.

# Our Solution

## A Unified Chat Assistant for Supply Chains

NexChain offers **a single chat assistant** that simplifies supply chain inquiries, automatically retrieving data across multiple sources to provide instant, clear answers with actionable insights.



# Key Features of NexChain

## 01 Natural Language

Users can ask supply chain questions in plain English.

## 02 Plain-Language Answers

Provides clear responses with rationale and recommended actions.

## 03 Audit Trail

Maintains detailed logs of every question and answer for compliance.

## 04 Automatic Routing

Directs inquiries to the appropriate data source automatically.

## 05 SLA Breach Detection

Automatically identifies and escalates potential SLA violations.

## 06 Query History

Offers past queries and suggested questions for new users.



# Building Blocks of NexChain

## 01 Chat Interface

Angular frontend ensures a responsive user experience.

## 02 AI Orchestration

Python FastAPI enables efficient AI task management.

## 03 Document Search

RAG with ChromaDB optimizes retrieval of SOPs and SLAs.

## 04 Secure Login

Java Spring Boot backend manages user authentication seamlessly.

## 05 Smart Routing

LangGraph framework directs questions to appropriate data sources.

## 06 Structured Answers

Text-to-SQL delivers accurate responses from PostgreSQL database.

# Database Design Overview

## 01 Centralized

A single PostgreSQL database serves as our core repository.

## 02 Core Tables

Essential tables include customers, orders, and inventory details.

## 03 Audit Trail

A comprehensive log tracks user actions and system interactions.

## 04 Security

Role-based access ensures data integrity and privacy compliance.

## 05 Tracking

We maintain tables for shipment and carrier tracking information.





# Backend Question Resolution Flow

## 01 User Interaction

User asks a question in the chat interface.

## 02 Intent Detection

LangGraph Supervisor agent detects intent and activates agents.

## 03 Response Compilation

Final Response Agent merges everything into one structured answer.

## 04 Authentication

Spring Boot authenticates the user and logs the request.

## 05 SLA Monitoring

Business Rule Agent checks for SLA breaches and delays.





# API Integration Overview

## **01      Safe Access**

MCP tools enable AI agents to access systems securely.

## **02      Standardized Tools**

Standard tools streamline external system connections for seamless operations.

## **03      API Reliability**

Failed calls retry automatically, enhancing user experience and trust.

## **04      Comprehensive Logging**

Every API call logs response time for traceability and analysis.

# Impact on Our Users

## **01 Supply Chain Manager**

Receives instant updates, enhancing response time to delays.

## **02 Operations Analyst**

Generates reports efficiently, eliminating the need for SQL.

## **03 Overall Efficiency**

Leads to faster operations and improved SLA compliance.

## **04 Customer Service Rep**

Answers customer inquiries quickly without data overload.

## **05 IT Auditor**

Traces data back to its source with a click.



# Future Advancements in NexChain

## 01 Integrations

Real integration with major platforms enhances operational efficiency.

## 02 Analytics

Predictive analytics will optimize supply chain delay management.

## 03 Dashboard

A control-tower dashboard provides live business intelligence insights.

## 04 Channels

Expanding to WhatsApp and voice channels for user convenience.

## 05 Notifications

Automated notifications via email and SMS keep users informed.

## 06 Support

Multi-language support will cater to global team needs.

Thank You

